

Variační odvození Einsteinových rovnic

→ funkcionál akce :

$$S = \int_{\Omega} \underbrace{(R - 2\Lambda)}_{\substack{\text{Lg invariantní} \\ \text{element} \\ \text{objektu}}} \sqrt{-g} d^4x + 16\pi \int_{\Omega} \mathcal{L}_{\text{mg}}(g^{\mu\nu}, g^{\mu\nu},_{;\nu}) \sqrt{-g} d^4x$$

- předáním fyzikl částí dostane rovnice fyzikl teorii jako KGE, EM, ...

→ uvažujeme záměnu variací $\delta g^{\mu\nu}$ s okrajovími podmínkami :

$$\delta g^{\mu\nu}|_{\partial\Omega} = 0 \quad \text{a} \quad \delta g^{\mu\nu}_{;\nu}|_{\partial\Omega} = 0$$

odpovídalo by $\Leftarrow$ není záležitost, proč by to tak mělo být

→ máme vztah $\frac{(\det M)_{,\lambda}}{\det M} = \text{Tr}(M^{-1} \cdot M_{,\lambda})$

$$\frac{\partial(-g)}{\partial g^{\mu\nu}} = (-g) g^{\alpha\beta} \frac{\partial g_{\alpha\beta}}{\partial g^{\mu\nu}} = (-g) g^{\mu\nu}$$

$$\frac{\partial \sqrt{-g}}{\partial g^{\mu\nu}} = \frac{1}{2\sqrt{-g}} (-g) g^{\mu\nu} = \frac{\sqrt{-g}}{2} g^{\mu\nu}$$

$$\delta \sqrt{-g} = \frac{\partial \sqrt{-g}}{\partial g^{\mu\nu}} \delta g^{\mu\nu} = \frac{\sqrt{-g}}{2} g^{\mu\nu} \delta g^{\mu\nu}$$

$$0 = \delta(g^{\mu\nu} g_{\mu\nu}) = g^{\mu\nu} \delta g_{\mu\nu} + g_{\mu\nu} \delta g^{\mu\nu}$$

$$\delta \sqrt{-g} = -\frac{\sqrt{-g}}{2} g_{\mu\nu} \delta g^{\mu\nu}$$

→ pro Ricciho skalar máme

$$R = g^{\mu\nu} R_{\mu\nu} \Rightarrow \delta R = R_{\mu\nu} \delta g^{\mu\nu} + g^{\mu\nu} \delta R_{\mu\nu}$$

$$R^{\mu}_{\mu\nu} = \Gamma^{\kappa}_{\nu\mu,\lambda} - \Gamma^{\kappa}_{\lambda\mu,\nu} + \Gamma^{\kappa}_{\lambda\nu}\Gamma^{\nu}_{\mu\kappa} - \Gamma^{\kappa}_{\nu\lambda}\Gamma^{\nu}_{\mu\kappa}$$

$$\delta R^{\mu}_{\mu\nu} = \delta \Gamma^{\kappa}_{\nu\mu,\lambda} - \delta \Gamma^{\kappa}_{\lambda\mu,\nu}$$

$$+ \delta \Gamma^{\kappa}_{\lambda\nu}\Gamma^{\nu}_{\mu\kappa} + \Gamma^{\kappa}_{\lambda\nu}\delta \Gamma^{\nu}_{\mu\kappa}$$

$$- \delta \Gamma^{\kappa}_{\nu\lambda}\Gamma^{\nu}_{\mu\kappa} - \Gamma^{\kappa}_{\nu\lambda}\delta \Gamma^{\nu}_{\mu\kappa}$$

$$\stackrel{\substack{\text{Palatiniho} \\ \text{identita}}}{=} (\delta \Gamma^{\kappa}_{\nu\mu})_{;\lambda} - (\delta \Gamma^{\kappa}_{\lambda\mu})_{;\nu}$$

→ kovariantní derivace dává smysl, protože $\delta \Gamma$ je vektor $\Leftarrow$ kovariantní část se počítá

$$\int g^{\mu\nu} \delta R_{\mu\nu} \sqrt{-g} d^4x = \int g^{\mu\nu} [(\delta \Gamma^{\kappa}_{\nu\mu})_{;\kappa} - (\delta \Gamma^{\kappa}_{\kappa\mu})_{;\nu}] \sqrt{-g} d^4x$$

$$= \int_{\Omega} \underbrace{[g^{\mu\nu} \delta \Gamma^{\kappa}_{\nu\mu} - g^{\mu\kappa} \delta \Gamma^{\nu}_{\nu\mu}]_{;\kappa}}_{\text{vektorová veličina}} \sqrt{-g} d^4x$$

$$[\cdot]_{;\kappa} = \frac{1}{\sqrt{-g}} [\sqrt{-g}(\cdot)]_{;\kappa}$$

$$= \int_{\Omega} [\sqrt{-g}(\cdot)]_{;\kappa} d^4x = \int_{\partial\Omega} \uparrow^{\kappa} n_{\kappa} d^3x = 0$$

$$\delta \Gamma^{\kappa}_{\alpha\Omega} = 0$$

→ overall dostaneme :

$$\delta S_g = \int_{\Omega} \left[\underbrace{R_{\mu\nu}}_{\substack{\uparrow \\ \text{variaci } g^{\mu\nu}}} - \frac{1}{2} \underbrace{R g_{\mu\nu} - \Lambda g_{\mu\nu}}_{\substack{\text{variaci } \sqrt{-g}}} \right] \delta g^{\mu\nu} \sqrt{-g} d^4x$$

→ pro nevyužití částí korekce :

$$\delta S_{\text{mg}} = 16\pi \int_{\Omega} \left[\frac{\partial(\mathcal{L}_{\text{mg}} \sqrt{-g})}{\partial g^{\mu\nu}} \delta g^{\mu\nu} + \underbrace{\frac{\partial(\mathcal{L}_{\text{mg}} \sqrt{-g})}{\partial g^{\mu\nu}_{;\nu}} \delta g^{\mu\nu}_{;\nu}}_{\substack{\text{okrajová část} \\ \text{využití}}} d^4x$$

$$\stackrel{!}{=} -8\pi \int_{\Omega} T_{\mu\nu} \sqrt{-g} \delta g^{\mu\nu} d^4x$$

$$T_{\mu\nu} := -\frac{2}{\sqrt{-g}} \left[\frac{\partial(\mathcal{L}_{\text{mg}} \sqrt{-g})}{\partial g^{\mu\nu}} - \left(\frac{\partial(\mathcal{L}_{\text{mg}} \sqrt{-g})}{\partial g^{\mu\nu}_{;\nu}} \right)_{;\nu} \right]$$

Gauge transformace

→ pro $x^{\mu} \rightarrow x^{\mu} + \xi^{\mu}$ se vektorový tenzor $T^{\mu\nu}$ o svoji Lieovu derivaci

$$g'^{\mu\nu} = g^{\mu\nu} - \xi^{\mu}_{;\nu} - \xi^{\nu}_{;\mu}$$

$$\delta g^{\mu\nu} := g'^{\mu\nu} - g^{\mu\nu} = -\xi^{\mu}_{;\nu} - \xi^{\nu}_{;\mu}$$

$$\int_{\Omega} T_{\mu\nu} \delta g^{\mu\nu} \sqrt{-g} d^4x = \int_{\Omega} T^{\mu\nu} (\xi^{\mu}_{;\nu} + \xi^{\nu}_{;\mu}) \sqrt{-g} d^4x$$

$$= 2 \int_{\Omega} \underbrace{T^{\mu\nu} \xi^{\mu}_{;\nu}}_{\substack{(\cancel{T^{\mu\nu}} \xi^{\mu})_{;\nu} - T^{\mu\nu}_{;\nu} \xi^{\mu} \\ = 0 \\ \text{3\mu je jako na konci} \\ \text{nebo více}}} \sqrt{-g} d^4x = -2 \int_{\Omega} T^{\mu\nu}_{;\nu} \xi^{\mu} \sqrt{-g} d^4x$$

$$\Rightarrow T^{\mu\nu}_{;\nu} \equiv 0$$

Palatiniho metoda

→ uvažujeme, že afinní konexe je nezávislá na metrice

- $\Gamma^{\kappa}_{\mu\nu}$ není dána Christoffelovými symboly
- $g^{\mu\nu}$ není kovariantně konstantní
- zájmeš nás gravitační část

$$\rightarrow \varepsilon_{\mu\nu\lambda} = \sqrt{-g} [\mu\nu\lambda]$$

$$\varepsilon_{\mu\nu\lambda;\rho} = \varepsilon_{\mu\nu\lambda,\rho} - \Gamma^{\nu}_{\rho\mu} \varepsilon_{\nu\lambda} - \Gamma^{\nu}_{\rho\nu} \varepsilon_{\mu\lambda}$$

$$- \Gamma^{\nu}_{\rho\lambda} \varepsilon_{\mu\nu} - \Gamma^{\nu}_{\rho\lambda} \varepsilon_{\mu\nu}$$

$$= (\sqrt{-g})_{;\rho} [\mu\nu\lambda] - \sqrt{-g} (\Gamma^{\nu}_{\rho\mu} [\nu\lambda] + \Gamma^{\nu}_{\rho\nu} [\mu\lambda] + \Gamma^{\nu}_{\rho\lambda} [\mu\nu])$$

$\downarrow$ aby to bylo nenulové, musí všechny indexy být různé $\Rightarrow$ ν je vždy ten index, co vypadl

$$= (\sqrt{-g})_{;\rho} [\mu\nu\lambda] - \sqrt{-g} \Gamma^{\nu}_{\rho\nu} [\mu\lambda]$$

$$\stackrel{!}{=} (\sqrt{-g})_{;\rho} [\mu\nu\lambda]$$

$$(\sqrt{-g})_{;\rho} = (\sqrt{-g})_{,\rho} - \Gamma^{\nu}_{\rho\nu} \sqrt{-g}$$

→ pro libovolný V^{μ} máme :

$$(V^{\mu} \sqrt{-g})_{;\mu} = V^{\mu}_{;\mu} \sqrt{-g} + \Gamma^{\mu}_{\mu\nu} V^{\nu} \sqrt{-g}$$

$$+ V^{\mu} (\sqrt{-g})_{;\mu} - V^{\mu} (\sqrt{-g}) \Gamma^{\nu}_{\mu\nu}$$

$$= (V^{\mu} \sqrt{-g})_{;\mu} + V^{\mu} \sqrt{-g} (\underbrace{\Gamma^{\nu}_{\mu\nu} - \Gamma^{\nu}_{\nu\mu}}_{\text{bez rozdílu}})$$

$$(V^{\mu} \sqrt{-g})_{;\mu} = (V^{\mu} \sqrt{-g})_{,\mu}$$

$\Rightarrow$ Gaussův vektor lze použít pro libovolnou afinní konexi

→ při variaci S_g je jediný člen u variace $R_{\mu\nu}$, protože pouze ten se variuje afinní konexi

$$\delta R_{\mu\nu} = (\delta \Gamma^{\kappa}_{\nu\mu})_{;\kappa} - (\delta \Gamma^{\kappa}_{\kappa\mu})_{;\nu}$$

$$\int_{\Omega} g^{\mu\nu} \delta R_{\mu\nu} \sqrt{-g} d^4x =$$

$$= \int_{\Omega} \underbrace{(g^{\mu\nu} \delta \Gamma^{\kappa}_{\nu\mu})_{;\kappa}}_{\text{Gauss}} \sqrt{-g} d^4x - \int_{\Omega} \delta \Gamma^{\kappa}_{\kappa\mu} (g^{\mu\nu} \sqrt{-g})_{;\nu} d^4x$$

$$- \int_{\Omega} \underbrace{(g^{\mu\nu} \delta \Gamma^{\kappa}_{\kappa\mu})_{;\nu}}_{\text{Gauss}} \sqrt{-g} d^4x + \int_{\Omega} \delta \Gamma^{\kappa}_{\kappa\mu} (g^{\mu\nu} \sqrt{-g})_{;\nu} d^4x$$

$$= \int_{\Omega} \delta \Gamma^{\kappa}_{\nu\mu} \underbrace{[\delta^{\nu\lambda} (g^{\mu\lambda} \sqrt{-g})_{;\lambda} - (g^{\mu\nu} \sqrt{-g})_{;\mu}]}_{\substack{\text{symetrická část v } \mu\nu \\ \stackrel{!}{=} 0}} d^4x$$

už je sym.

$$\frac{1}{2} \delta^{\nu\lambda} (g^{\mu\lambda} \sqrt{-g})_{;\lambda} + \frac{1}{2} \delta^{\mu\lambda} (g^{\nu\lambda} \sqrt{-g})_{;\lambda} - (g^{\mu\nu} \sqrt{-g})_{;\mu} \stackrel{!}{=} 0 \quad \left| \begin{matrix} \mu \\ \kappa \end{matrix} \right.$$

$$\frac{1}{2} (g^{\nu\lambda} \sqrt{-g})_{;\lambda} + 2 (g^{\nu\lambda} \sqrt{-g})_{;\lambda} - (g^{\mu\nu} \sqrt{-g})_{;\mu}$$

$$= \frac{3}{2} (g^{\nu\lambda} \sqrt{-g})_{;\lambda} \stackrel{!}{=} 0$$

$\Rightarrow$ dosazením do první rovnice

$$(g^{\mu\nu} \sqrt{-g})_{;\mu} = 0$$

$$0 = (g^{\mu\nu} \sqrt{-g})_{;\mu} = g^{\mu\nu}_{;\mu} \sqrt{-g} + g^{\mu\nu} (\sqrt{-g})_{;\mu} \quad | g_{\mu\nu} (*)$$

$$= \frac{\sqrt{-g}}{2} g^{\alpha\beta} g_{\alpha\beta;\mu}$$

$$g_{\mu\nu} g^{\mu\nu}_{;\mu} + \frac{1}{2} 4 g^{\alpha\beta} g_{\alpha\beta;\mu} \stackrel{!}{=} 0$$

$$g_{\mu\nu} g^{\mu\nu}_{;\mu} - 2 g^{\alpha\beta} g_{\alpha\beta;\mu} \stackrel{!}{=} 0$$

$$g_{\mu\nu} g^{\mu\nu}_{;\mu} \stackrel{!}{=} 0$$

$\Rightarrow$ dosazením do (*)

$$g^{\mu\nu}_{;\mu} \stackrel{!}{=} 0 \quad \text{Levi-Civita konexe}$$